Modeling Customer Churn to Quantify Retention ROI

Predicting churn, churn drivers, and revenue uplift from customer retention strategies

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Tools used: Python
lifelines, XGBoost, Survival Analysis

Data Acquired from [Maven]
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Why Churn Matters

New customer acquisition is more costly than customer retention, and so retention strategies are key for businesses. But, to be able to ideate these strategies, we need to understand: **when our customers tend to exit our offering (i.e., churn) and why.**

If we analyse our customer base, we can understand what **characteristics make them more or less susceptible to churning**. We can leverage this understanding to quantify the monetary effect of our strategy (our ROI), and to rank strategies by priority.

Goals of the Analysis

In this fictional dataset from [Maven Analytics], we are given information on **telecom customers**, including demographics, tenure, and services used.

We aim to use this data to answer the below questions:

- Will this customer churn? (churn **prediction**)
- Which **variables** contribute most to churn? Which are **risk or protective** factors?
- What is the **probability** our customers have not yet churned at X months?
- If we engaged in interventions, what would be the **uplift to our customers' baseline tenure**? What would be our expected **incremental revenue**?

Overall, by understanding when and why our customers churn, we can formulate targeted retention strategies with a clearly defined ROI estimation

Spoiler alert!

By the end of this analysis, we predict:

When stratified by contract-type (a structural barrier to churn), the Cox Proportional Hazards model tells us that social factors are statistically significant predictors of churn.

Within the same contract type, each additional **referral reduces churn hazard by 26%**, and each additional **dependent reduces churn hazard by 33%**.

If we ran a **referral incentive program for our month-to-month contract customers**, and expected that **30% of this cohort participated,** we would see:

- A baseline tenure uplift of **5.5 months per treated customer**, translating to 0.85 months per customer across the entire portfolio.

- Overall incremental revenue of **$386k** from increased tenure only, not yet taking into account discounting or new customer gains.

Predicting Churn

- XGBoost
- Feature Importance

XGBoost

Data Preparation

- **Import and read data**
  Data source: Maven
  38 columns x 7043 rows
- Check for non-null values
- Define X (predictor variables) and Y (Churned/Not Churned)

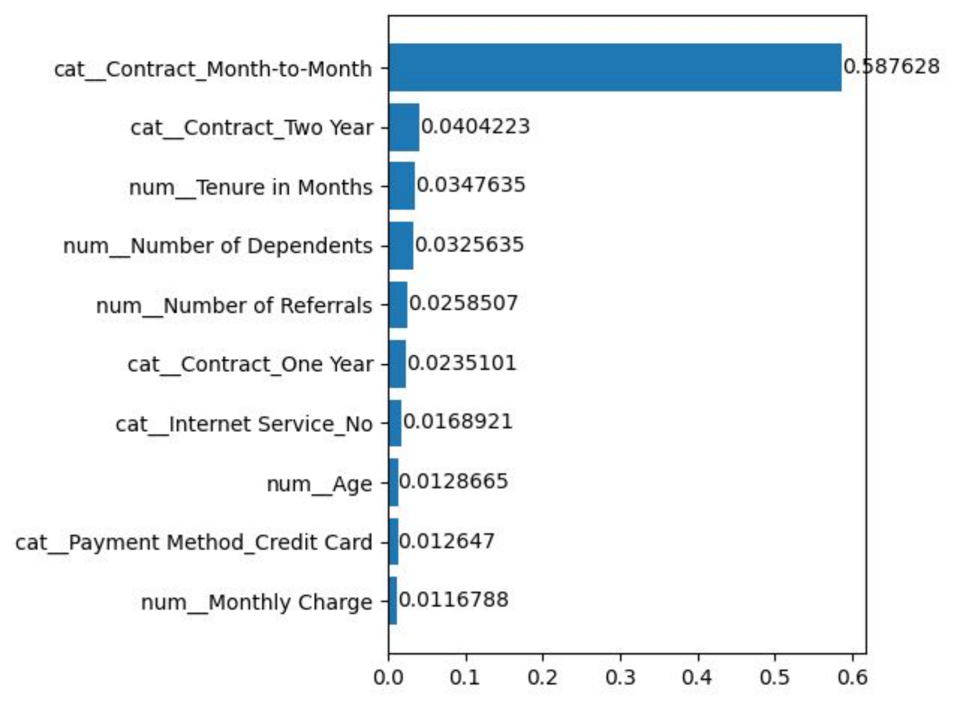
Data Preprocessors

- **Cull non-essential columns** (redundant, high cardinality)
- **Define Numerical and Categorical Columns**
  - Num: dtype int64, float64
  - Cat: dtype object
- **Define Preprocessors**
  - Pipeline of Imputers and Encoders

XGBoost

- **Define Train Test Split**
- **Define XGBClassifier**
  - n_estimators, learning_rate, max_depth, eval_metric, early_stopping_rounds
- **Score Model**
  - Mean Accuracy: **0.89**

Feature Importance *



Contract Type **

A strong indicator of churn, with a month-to-month contract being particularly susceptible

Tenure in Months

The longer a customer has remained a customer, the stickier they become.

Number of Dependents & Referrals

The more involved a customer’s social network is in their telecom service, the more likely they are to stay

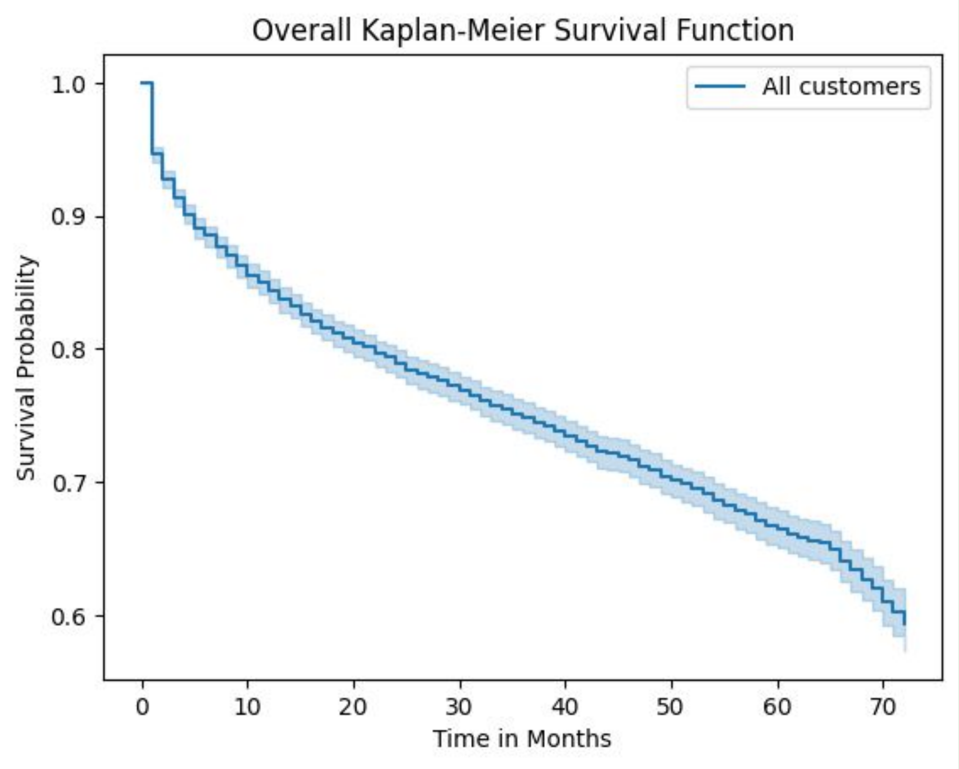
* Feature importance provides magnitude but not direction. We make some inferences here.

** Contracts can be considered a structural barrier to churn rather than a predictor. We will address this later.

Survival Analysis

- ➢ Kaplan-Meier Curves
- ➢ Cox Proportional Hazards
- ➢ Tenure Uplift Example

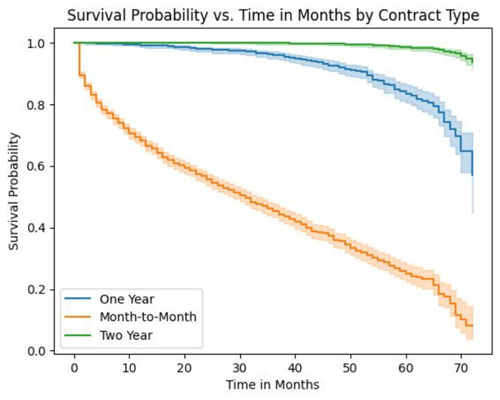
Kaplan-Meier Curves



Whole Portfolio

- **We expect more than half our customers to remain past 70 months**
- This view alone is not very informative, so we'll move to: **what is the survival probability of different cohorts of customers over time?**
    - Note that the Kaplan-Meier analysis **cannot model multiple covariates** at the same time, so we must look at each variable separately

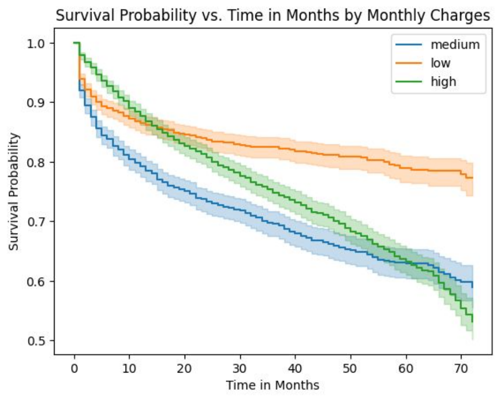
Kaplan-Meier Curves



Contract Type

- In Feature Importance, contract type was **highly predictive** of churn
- **Longer contracts** see the highest customer survival over time

- Note that, assuming these contracts are lock-in, contract type would be considered a **structural barrier** to churn rather than a predictor



Monthly Charges

- Customers with **low monthly charges** are more likely to remain over time

- To confirm that the difference among the curves is statistically significant, we ran a K-sample log rank test (p-value < 0.005)

Cox Proportional Hazards

A CoxPH model will allow us to predict **both the magnitude and direction** of effect on churn **among multiple covariates simultaneously**, producing Hazard Ratios.

In this example, we ran only select columns of our dataset to maintain focus on important features.

Sample Dataset

Churn Event: Customer Status
Event Duration: Tenure in Months
Dataset: Contract Type, Monthly Charge

Baseline contract type: monthly

	Hazard Ratio	p-value	
Monthly Charge	1.00	0.14	No statistically significant effect to churn when evaluated in conjunction with contract type
Contract_One Year	0.10	<0.005	**90% less likely to churn** compared to monthly contract
Contract_Two Year	0.01	<0.005	**99% less likely to churn** compared to monthly contract

Hazard Ratio: 1 = No effect, <1 = Protective, >1 = Risk Factor

Stratified Cox Proportional Hazards

With Contract type skewing our earlier results due to being a barrier to churn rather than a predictor, we instead **stratify our dataset by Contract type.**

We now see hazard ratios **within** each contract stratum.

Sample Dataset

Churn Event: Customer Status
Event Duration: Tenure in Months
Dataset: Age, Number of Referrals, Number of Dependents, Monthly Charge

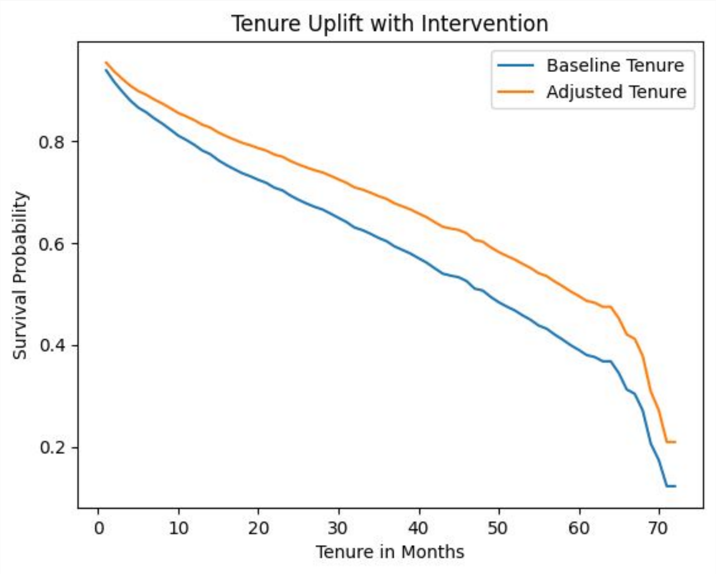
Stratified by: Contract Type

	Hazard Ratio	p-value	
Age	1.01	<0.005	1% more likely to churn per year aged
Number of Referrals	0.74	<0.005	**26% less likely to churn** per referral
Number of Dependents	0.67	<0.005	**33% less likely to churn** per dependent
Monthly Charge	1.00	1.00	No statistically significant effect to churn

Retention Intervention

What if we did a referral campaign aimed at our month-to-month customers?

For this exercise, we will assume that **30%** of our month-to-month customers participated in the campaign, increasing their number of **referrals by +1 each.** We will translate **reduced churn hazard** (26% reduction) into an **increase in tenure**, then consequently an **increase in revenue.**



RESULTS

+ 5.54	Months	Avg tenure uplift per treated customer
+ 0.85	Months	Avg tenure uplift per customer across entire portfolio
$ 386k	Revenue	Overall incremental revenue gained from the intervention
$ 54.89	Revenue	Avg incremental revenue per customer across entire portfolio

What Next?

- Model Validation Framework

Model Validation Framework

3 Months	Short Term *Churn sense check*	• **Compare churn rate** post-intervention in control vs treatment (intervention) groups -- expected reduction of 26%
6 Months	Medium Term *Survival separation*	• **Compare Kaplan-Meier curves** for control vs treatment groups post-intervention ○ Confirm statistical significant difference via log rank test • **Revisit hazard ratio** via CoxPH, using treatment as a variable ○ Does treatment reduce hazard as originally expected (26%)?
12 Months	Long Term *Tenure uplift confirmation*	• **Revisit baseline remaining tenure** for both control vs treatment groups separately -- If a difference exists, does it match our prediction (5 months)?

Thank you!